SoK: Agent Communication Side Channels Feedback Paths and Executed Cover

***Abstract*—Agent communication carries task information across services, while privacy decoys can become business operations. We systematize these two execution boundaries using fifteen direct Agent studies, forty-four settings and twenty-two established traffic-analysis anchors. A transfer matrix and deployment maps connect each reported inference to its observer, workload and protection conditions. The synthesis identifies two design requirements: evaluate the model and recipient views left after a tool-side intervention, and measure business-state integrity alongside task completion when cover executes. A six-condition study with two fixed-weight LLM planners makes the second requirement concrete. Across 3,600 synthetic executions over MCP Streamable HTTP and TLS 1.3/HTTP/2, live-write cover completed 102/120 and 76/120 held-out tasks, yet preserved the intended global state in only 7/120 and 3/120. Both state-integrity reductions survived the prespecified multiplicity correction. The same runs retain substantial traffic recognition, and the visible-decoy comparison remains inconclusive. These measurements complement the published executed-decoy utility gap; model-view and adaptation evidence remains partly preprint-dependent. Together, the evidence map and controlled case show why privacy, task completion and collateral state changes must be evaluated under the same intervention.**

1. Introduction

An Agent receives the instruction *process the next support ticket*. It reads a request, checks an order and approval, consults a model, and submits a refund. Encryption hides messages from a network observer but leaves contacted services, sizes and timing visible. The tool's answer also becomes input to another service: the model. A privacy decoy can cross a second boundary if it invokes a business tool. The requested refund may succeed while an unrelated order or shared quota changes. Figure 1 follows these two boundaries.

Protection follows the execution across services and state. Classical traffic analysis supplies observer models, segmentation and adaptive evaluation. We connect twenty-two mechanism anchors to fifteen direct Agent studies and forty-four settings to determine which conditions carry over and which need an application-level constraint. The resulting SoK centers on two findings.

Feedback leaves separately observable representations. Research queries can be recovered from tool destinations, and task labels can be recognized on model connections [2], [9], [4]. A contacted tool additionally sees its arguments. These observations place protection at different boundaries: tool-side padding leaves later model requests outside its scope, while transit encryption leaves the recipient's arguments visible. Our map connects each inference to the rights and execution stage that make it possible. A paired forwarding pilot measures first-feedback and complete-execution exposure; its adjusted primary comparisons remain inconclusive.

Task completion can conceal executed-cover damage. Cover discarded at a transport endpoint consumes bandwidth; cover executed by an Agent also changes application work. WRA's five decoys perform additional research and lower query recovery, but its task-utility scores belong to a separate visibility intervention [2]. This leaves a concrete evaluation question: what happens to the original task and surrounding state on the protected runs?

We test that question with Qwen3:8B and Llama3.1:8B selecting actual tool actions through LangGraph and MCP. Six paired conditions compare no cover, proxy discard, read-only cover, isolated execution, live-state writes and isolated responses visible to the planner. Under the constructed live-write policy, 95/120 Qwen and 73/120 Llama test executions completed the requested task while leaving incorrect global state. Without cover, those counts were 0/120 and 13/120. Both state-integrity reductions survive multiplicity correction; only Qwen's completion reduction does. The result identifies what a completion-only evaluation misses, within a controlled adverse-write design.

Website fingerprinting (WF) and machine-learning security already establish realistic populations, segmentation, transfer and adaptive evaluation [15], [48], [34], [16], [55], [19]. Our contributions are their *transfer and deployment maps* for Agent execution, a *source-bound account of remaining views and executed cover*, and a *matched measurement of inference, task completion and state integrity*. Among four published or preprint query/workflow privacy settings from two sources, adaptation is located in two and original-task utility in one; none documents both. The new controlled study supplies all three endpoints for its own workload and keeps its results separate from this literature count.

2. Scope and Evidence Construction

The evidence map connects Agent execution to an observed communication signal, a hidden target and the rights needed to observe it.

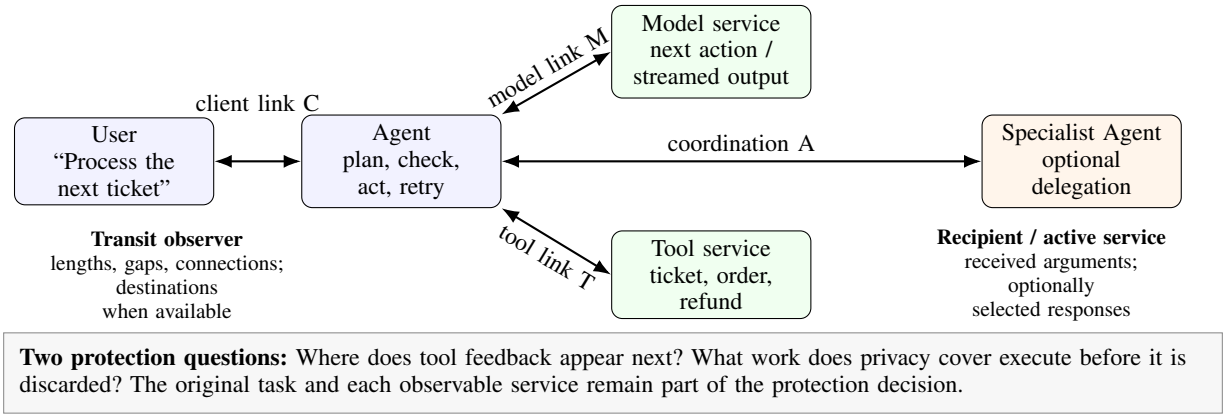


Figure 1. Running support-ticket example and observation rights. A tool result returns to the Agent and can enter the next model request. Each encrypted link exposes its own traffic shape; a recipient additionally sees the request addressed to it. Concurrent tickets may share a connection on either link. The arrows represent application communication, not a one-to-one mapping from tool events to packets.

2.1. Inclusion and retrieval

An Agent uses an LLM to select actions, with external feedback able to affect later action or termination. Client, tool, model and inter-agent interfaces are eligible. Transit observers obtain traffic metadata; contacted recipients see their addressed requests; active auditors also choose probes or challenges. Negative results qualify: PrivacyTrace's metadata/order condition remains included despite scoring below its prior reference [37]. Whole semantic trajectories or browser instrumentation alone enter the adjacent layer, while deliberate covert messages and unrelated prompt-injection defenses fall outside this question [38], [27].

The fixed eligibility window covers first public submissions from 1 January 2024 through 9 November 2026. Twelve topic queries, a native arXiv query and inclusion rules preceded the retrieval pass; title/publication checks and one-step citation follow-ups completed it. The arXiv query returned 180 records. Public-web queries supplied 33 scholarly candidates across 101 displayed URLs; deduplication yielded 209 records. Two citation leads brought screening to 211: fourteen direct, forty adjacent, 155 outside scope and two unresolved identities. The eligible CHeaT (Cloak, Honey, Trap) seed adds the fifteenth direct source (Figure 2). The SSH honeypot entered through citation tracing, illustrating the value of combining retrieval routes. Exact queries and dispositions are in the supplement.

Twenty-two separately selected mechanism anchors comprise sixteen published communication studies and six serving studies. They supply the established relations in the transfer matrix; the fifteen direct studies supply the Agent application evidence. These layers retain separate inclusion records and denominators. The anchor set is selected for mechanism correspondence, rather than a census of classical traffic analysis.

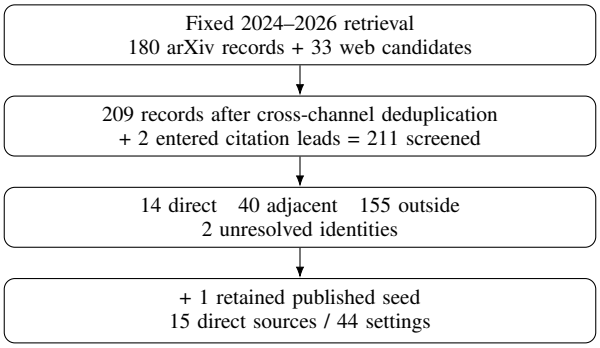


Figure 2. Executed coverage audit. Counts are deduplicated work/lead records; two identities remain unresolved. Mechanism anchors use a separate inclusion rule.

2.2. Extraction and source dependencies

Forty-four setting records bind source version and locator to access, features, learner, target, population, split, intervention, result and cost. Labels, execution outcomes, failure treatment and policy context are recorded alongside them. WRA and CHeaT are checked against official USENIX texts. One HTTP study has a verified ACM workshop identity, while its detailed measurements come from the extended arXiv version [52], [53]. Twelve other works lack independently verified formal publication identities at the cutoff.

Text reading, displayed-number reconstruction, selected-code inspection and data recomputation have separate entries in the extraction sheet. For example, WRA's decoy and visibility conditions are checked in the published text; OHTTP refitting is checked in selected code. Each check supports its matched premise. Section 9.1 reports what remains when sources are removed. Extraction is author-performed with AI assistance. A coding-manual pilot compares the released author coding with two non-blind AI passes from the same model. Structural fields agree on all forty-four rows, while execution-outcome definitions produce substantial disagree-

ment. Supplement S4.1 gives the original encodings, descriptive agreement and source adjudication; this is not independent human validation.

For the descriptive summary, every setting receives an access class and intervention purpose. Protected-condition adaptation and original-task utility are counted within the four query/workflow privacy settings. Three identity-obfuscation settings and two adversarial-Agent blocking settings have different objectives; thirty-five settings contain no protection comparison. A located item identifies evidence in the retained record. An unlocated item leaves that question open. The supplementary reporting ledger lists the assignments and source locators.

2.3. Position among security surveys

Broader surveys organize LLM and Agent threats by lifecycle, attack surface and defensive role [56], [57], [58]. We connect communication measurements to execution, access and protection cost. The supplementary deployment section gives the detailed scope comparison.

3. Reading the Evidence Map

The same ticket can expose different facts at the network link, the tool recipient and the model service.

3.1. From private task to observed execution

An execution maps a hidden task or state through an agent policy and environment into communication, then into an observer's features:

$$\text{hidden task/state} \longrightarrow \text{agent execution} \longrightarrow \text{observed communication} \longrightarrow \text{inference}.$$

Different cuts through this chain expose different objects. A client link can reveal response mode or hosted-agent identity. Tool egress exposes contacted services and request/response shape. The model link carries prompts, tool feedback and model responses. Inter-agent links expose delegation and coordination. Figure 1 places these observations on the feedback path.

3.2. What the inferred label means

The ticket example separates four labels. *Intent* is the requested objective or assigned task family. *Execution* is the recorded action sequence. *Success* states whether that sequence and its final state meet the task's criterion. *Authorization* states whether the action is permitted for the principal under the applicable policy. A failed refund task can still contain a refund attempt; a successful refund can still be unauthorized. An evaluation should bind its target to the corresponding reference instead of letting one label stand for all four.

Exposure is relative to both the target and the observer. The same execution may expose intended-task information on M and actual contacted services on T. Initial instructions, tool feedback and later decisions all contribute to the complete M trace. Assigning a task-family label to that trace measures intention recognition; attributing the signal to a particular feedback step requires a further controlled comparison.

3.3. What each record specifies

Six fields give a score its meaning: the *interface and visible representation*; *observer access*; *prior information* supplied by the evaluator; *target*; *population and base rate*; and *protected training access*. These fields define the comparison record. The matrix records the fields once; the reporting checklist in Section 8 turns them into entries for future studies.

Access determines what a feature can mean. A server can inspect decrypted HTTP/2 properties that a transit observer cannot. The content-aware MCP study's "metadata" includes argument hashes, and its graph uses content matches [45, Secs. 4.1–4.2]. Its representation therefore requires more than packet metadata. Logs and browser instrumentation can provide evaluation labels without becoming attacker inputs.

ANT's supplied stage boundary is prior information; its 2,101 stages and 47 labels define the decision population; the primitive is the target [9]. In the live OHTTP comparison, an unchanged random-forest (RF) family refits on protected data, so it is an adapted attacker [3]. These examples show how to read access, target, population, supplied boundaries and training conditions in sequence. A source setting that needs an unrepresented field would require revising the map.

3.4. Deployment determines who can see each signal

Table 1 maps common deployments to observation rights and the corresponding source evidence. A visible model link is an access condition, not a property granted to every attacker. Captured packets, TLS records, HTTP streams, server-sent events (SSE) and tool calls are different units. Segmentation, buffering and retransmission change their correspondence. Supplement S7 gives the protocol-level map.

Two simultaneous tickets may use separate connections or share an encrypted HTTP/2 connection. In the latter case, a transit observer sees combined packets while stream identifiers remain encrypted. A gateway that buffers SSE output changes the pattern again. The connection and flush policy therefore determine the measured trace.

The same reasoning explains defense placement. Padding a tool message before a gateway does not specify the gateway's output. Local MCP over `stdio` removes that network tool leg, while remote model requests may remain. Shared key/value (KV) cache state creates an additional timing interface for a querying tenant. The map therefore records the transport, observation layer, terminating parties

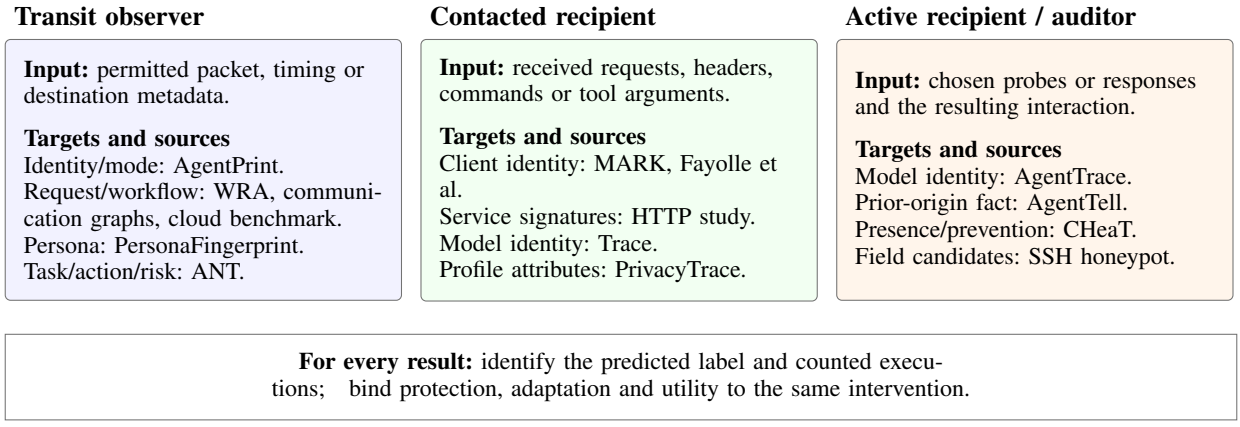


Figure 3. Evidence space by observation rights. The fifteen studies are placed by the mapped settings, not ranked by accuracy or access strength. Some offer richer additional views; Tables 3–4 identify those conditions and their metrics. Known-service signatures and field candidates are descriptive endpoints, not held-out recognition rates.

TABLE 1. DEPLOYMENT AND ACCESS MAP. THE CONFIGURATIONS ARE ANALYST-DEFINED; CITED SETTINGS TEST THE LISTED VIEW, NOT EVERY COMPONENT OF THE DEPLOYMENT. A GATEWAY'S PLAINTEXT ACCESS REQUIRES TLS TERMINATION.

Deployment	Observer and permitted view	Existing evidence for that view	Protection consequence
Enterprise egress	A transit monitor sees routed model/tool flows, sizes, times and any exposed destination association.	WRA: supplied research-domain traces; ANT and cloud benchmark: model/combined traffic [2], [9], [4].	Specify which services cross the monitored egress. A union of flows is available only when the monitor can observe them.
Shared gateway or relay	Transit sees multiplexed gateway output. A terminating gateway can additionally read addressed requests.	Live OHTTP: passive-link inference under stated relay trust [3]; PrivacyTrace: semantic recipient/gateway access [37].	Separate relay association protection from content trust. Connection sharing changes the observation unit.
Local model, remote tools	An external observer sees remote tool traffic; the local model call has no corresponding external network leg.	WRA's destination-recovery mechanism is relevant to tool egress [2]; whole-deployment protection is untested in the mapped comparison.	Evaluate tool and downstream calls. Do not import a remote-model attack score into an absent network leg.
Cloud model, remote tools	An authorized vantage may see either or both encrypted legs. A contacted tool sees its own arguments.	ANT/cloud benchmark test model traffic [9], [4]; PrivacyTrace tests recipient-visible representations [37].	Check permitted model and tool views after the intervention; recipients and passive monitors need different protections.

and shared state alongside each result. Section 7.2 compares real HTTP/2 concurrency with paired feedback and shaping interventions.

4. Classical Conditions in Agent Execution

Established traffic-analysis results carry conditions on access, observation units, workloads and interventions. Table 2 connects those conditions to the Agent evidence. Its central distinction is between an evaluation principle that carries over and an empirical performance result that requires a matched Agent test.

4.1. Reading the transfer matrix

Open-world prevalence, false alarms, adaptive attacks and workload shift are established concerns [15], [49], [34], [16], [55], [19]. Localization also has direct precedents: Wang and Goldberg study traffic splitting without reliable page boundaries; BAPM and ARES address mixed multi-tab traces [48], [46], [47]. WF is neither inherently deterministic nor restricted to one link. Here these issues interact with feedback across independently controlled services and cover executed as application work.

Retained means that an established evaluation obligation still applies; *extended* identifies an additional Agent execution condition; *access mismatch* prevents importing a result under the proposed observer; and *untested* means that the mapped evidence leaves the specific performance claim open. These relationships are separate from the number and publication status of supporting sources. Distributed tracing illustrates access mismatch: WAP5 and vPath use host instrumentation [32], [33], which can label an experiment without becoming a passive attacker's input.

TABLE 2. TRANSFER OF ESTABLISHED CONDITIONS TO AGENT EXECUTION. RELATIONSHIP LABELS CONCERN THE STATED MAPPING; EVIDENCE STRENGTH AND MISSING TESTS APPEAR SEPARATELY. A MISSING AGENT TEST DOES NOT REFUTE THE CLASSICAL RESULT.

Established relation and condition	Agent mapping / consequence	Relation	Agent evidence and remaining test
Unknown boundaries and background affect event recognition [48], [46], [47], [35].	Tool bursts, model waits and task intervals are different units. Evaluate proposed events with false alarms as well as classification.	Retained	ANT supplies stage boundaries; the local concurrent study tests particular detectors. Greater Agent localization difficulty is untested.
Defense evaluation must allow adaptation to the protected distribution [34], [16], [19].	Record protected training access and still-visible links; an unchanged learner family can refit.	Retained	Transformed and live OHTTP settings report adaptation in one preprint [3]; WRA's matched protected examples remain unresolved.
Coarse timing/volume can remain useful under specified padding [34]; serving streams expose length/gap signals [20], [12].	Name the shaping policy, receiving hop and Agent workload before carrying over a measured effect.	Untested generalization	Cloud and ANT measure model-view recognition [4], [9]. A general protected-Agent SSE claim is not tested by these observations.
An observation result is tied to the visible paths and representations [30], [36].	Feedback forwards tool information into later model requests across separately controlled services; evaluate every permitted residual view.	Extended	Published WRA tool-side evidence and preprint model/recipient results establish different exposed interfaces. Causal feedback attribution needs its own test.
STP tunnel endpoints remove cover before it reaches devices or servers [36, Sec. 7.1].	When cover executes as research, add original-task resource, state and feedback constraints to the traffic budget.	Extended	Published WRA reports executed research decoys and utility under a different intervention [2]. Our controlled LLM case separates task completion from state integrity (Section 7.1); it does not evaluate WRA writes.
Host traces can reconstruct distributed request paths [32], [33].	Instrumentation can provide ground truth; a transit attacker must infer from its own permitted observations.	Access mismatch	ANT stage labels and local task intervals are evaluator-provided. Host tracing alone cannot establish passive encrypted-stream localization.

Five questions locate the resulting evidence: signal mechanism, supplied-unit recognition, named deployment transfer, stream localization and protection. They are distinct tests. A reported score binds a target to an observation and a tested deployment; Tables 3–4 preserve those bindings across the field.

4.2. Task boundaries and execution changes

AgentPrint supplies browser interactions, WRA research traces, the communication-graph study workflows, the cloud benchmark execution timestamps, and ANT runtime-defined stages [1], [2], [3], [4], [9]. ANT's 47-group primitive test scores 0.355–0.462 accuracy on 2,101 supplied segments. Its episode split separates parent executions, while the supplied boundaries define a recognition test. Event localization additionally requires recall and false events on proposed intervals.

PersonaFingerprint keeps one session's windows in one partition [5]. A neighboring published study classifies non-overlapping 5,000-packet windows without page markers, primarily testing human browsing with Agent-generated training traces [43]. PINGPONG and voice-invocation work demonstrate detection against background traffic [35], [22].

Transfer names what changes while the target remains fixed: wording, session, composition, topology, scaffold or model. Reported composition, topology and scaffold losses concern different axes (Section 5.3). Simultaneous Agent tasks and model waits also change how tool bursts relate to complete executions. Our live comparison separates supplied-window inference from localization and retains physical connection identities.

5. Two Execution Dependencies for Protection

The field map connects two properties of Agent execution to defense design: information passes between services, and cover can become application work. Table 5 states the observation conditions under which particular signals matter.

5.1. Feedback joins separately observable services

The support-ticket execution exposes a selected tool destination, arguments received by that tool, a later model request and the model's response. WRA recovers query meaning from contacted domains under its name-association and supplied-trace conditions [2]. ANT, a network-traffic benchmark for Agent behavior, and the cloud benchmark measure task recognition from model traffic [9], [4]. The cloud benchmark reports ten-class balanced accuracy .853/.860 for graph/star coordination with all 247 features, and .540/.587

TABLE 3. ATTACK FIELD MAP, PANEL A: TRANSIT OBSERVATIONS. SCORES RETAIN THEIR METRIC, TASK AND SUPPLIED UNIT; THEY ARE REPRESENTATIVE REPORTED OPERATING POINTS, NOT A CROSS-PAPER RANKING. THE MATRIX RECORDS ALL FORTY-FOUR SETTINGS. P: INSPECTED PREPRINT; V: INSPECTED PUBLISHER VERSION.

Interface / observer	Work / target	Observable features / learner	Evaluation unit	Reported result and boundary
Client–agent; transit	AgentPrint [P] [1]; hosted agent	Directional packet counts/bytes in 1,800 relative-time bins; CNN	Supplied browser interaction	Mixed-flow macro-F1: .8660 closed, .8477 open; 50 monitored agents. Behavior-F1 conflict is kept separate.
Agent–Web; transit	WRA [V] [2]; research query	Contacted domains and timing; ICL or fine-tuned LLM	Supplied research trace	Functional query recovery .770 with domains; .725 with domains plus timing in the compared visibility views. This is a reconstruction score.
Inter-agent / tools; transit	Communication graphs [P] [3]; workflow	Message/graph, size and timing descriptors; random forest	Workflow record	Nine-class accuracy .67; within-label composition holdout .18. The live defense test instead has six classes.
Agent–model; transit	Cloud benchmark [P] [4]; assigned task	247 traffic descriptors; random forest	Logged successful run	Ten-class balanced accuracy .853/.860 for graph/star coordination. Size-excluded feature protocols score .540/.587.
Agent–Web; transit	PersonaFingerprint [P] [5]; browsing persona	Signed lengths/interarrival times; CNN representation and prediction heads	Fixed packet window; session split	Per-site macro-F1 about .78–.91; pooled-summary discrepancy excluded from synthesis. Explicit other-class tests have a different population.
Combined egress, M or remaining links; transit	ANT [P] [9]; task, primitive, risk	Direction/length/time; 13 learners including DF and TikTok	Supplied episode or logged stage	Five-task accuracy .431–.727; 47-primitive accuracy .355–.462. Risk test: 20 positive/189 negative, requiring recall and false alarms.

when size features are excluded and 100 of the 134 remaining features are selected. Both feature availability and selection change in that comparison. ANT's model-view values reconstructed from rounded figures remain approximate quantities in Supplement S2.

Recipient access changes the inference problem. PrivacyTrace's tool metadata/order yields attribute-inference ASR 14.53%, below its 17.97% schema-prior reference; adding arguments yields 27.71% [37]. MARK's combined F1 .971 also uses browser behavior, while its protocol/header features are available at the contacted website [6]. Encryption against transit observation does not hide an addressed request from its recipient.

These results identify where task information is observable. They lead to a testable question: with the initial instruction fixed and task-bearing feedback released later, how much recognition remains on the model link under a stated tool-side policy? Section 7 tests this question with constructed schemas and reports both the first-feedback and complete-execution views. Initial instructions, retained task fields and model responses are distinct possible contributors to the latter.

Defense consequence. Enumerate the services and representations available to the attacker, then evaluate the views remaining after an intervention. A joint observer can use a single-link decision rule, even when a particular fitted joint classifier scores lower. At the application boundary, minimize forwarded context while preserving the information needed for the task; at the transport boundary, evaluate visible sizes, timing and association. Their effects belong to separate contrasts.

5.2. Cover termination determines the execution constraint

Transport cover can terminate before business execution. Executed cover performs work and may affect resources, state or later decisions. The difference determines which utility must accompany a privacy measurement.

WRA's five research decoys reduce functional query recovery from .770 to .565 [2, Sec. 8.1]. Its utility scores 8.55 and 8.54 instead compare 100% and 10% destination visibility. Thus the utility observation does not describe the decoy intervention that produced the recovery reduction. WRA establishes additional research execution; it does not report harmful business-state writes. Matched protected-condition attacker examples also remain unresolved in the selected evidence.

The live OHTTP study refits a random forest on each protected capture and pairs six-class accuracy .30 with approximately thirteen-fold bytes and eleven-fold latency [3, Sec. 9.7]. Its separate nine-class .67 score belongs to another comparison. This design point binds adapted inference to transport cost; matched original-task utility was not located.

Task completion is only one execution outcome. The LLM-planned study in Section 7.1 measures inference, requested-task success and exact business state on the same runs. Live writes leave 95 and 73 completed tasks with collateral state errors in the two 120-fixture test populations. This supplies a controlled example of the

TABLE 4. ATTACK FIELD MAP, PANEL B: CONTACTED RECIPIENTS AND ACTIVE AUDITORS. THESE OBSERVERS SEE REQUESTS ADDRESSED TO THEM OR ISSUE PROBES; THEIR RESULTS DO NOT MEASURE ENCRYPTED-TRANSIT INFERENCE. E: INSPECTED EXTENDED PREPRINT WITH A SEPARATELY VERIFIED PUBLISHED SHORT VERSION. NR: NO RECOGNITION RATE REPORTED IN THE MAPPED SETTING. ASR: ATTACK SUCCESS RATE.

Interface / access	Work / target	Observable features / learner	Reported result and evaluation boundary
Web recipient	MARK [P] [6]; client identity	TLS/HTTP2 and headers; decision tree	F1 .725 for TLS/HTTP2, .799 for headers; .971 combined with browser instrumentation, a richer view. First-pass trials.
Web recipient	Fayolle et al. [P] [7]; client / access	Protocol signatures; richer behavior classifier and site challenges	Fingerprint analysis excludes failed visits. Access success is separately verified through form/cookie/page identifiers; not a privacy score.
Documentation recipient	HTTP signatures [E] [52]; service/client	Request method/version, headers and origin; descriptive comparison	Nine coding agents and six assistants, three trials each; NR. Request signatures are observations, not a held-out identification rate.
Terminal recipient	Trace [P] [51]; model family	Command TF-IDF unigrams/bigrams; linear SVM	Seven families; 2,028 clean/injected sessions: macro-F1 .981; held-out scaffold .815. A separate proprietary-scaffold test reports 78% accuracy.
Tool interface; active audit	AgentTrace [P] [50]; model identity	Serialized arguments, parser errors and repair order; Markov likelihood	Six models, 50 held-out probes per trial: 96.0% accuracy; 10 probes: 92.8%. Repeated stochastic trials reuse probe identities.
Web recipient; active choice	AgentTell [P] [8]; prior-origin fact	Which offered action is requested; loaded/cold comparison	Matching choice in 61.1% of 7,630 loaded sessions; the leakage measure subtracts cold matching. Actions share a completable probe task.
Tool recipient / gateway	PrivacyTrace [P] [37]; profile attribute	Tool metadata/order; content-added LLM inference	Metadata ASR 14.53%, schema prior 17.97%; arguments 27.71%. 68,000 attribute slots are clustered within 4,000 trajectories.
Contacted asset; active challenge	CHeaT [V] [10]; agent presence	Planted URL following; separate password-output control	Presence-detection mechanisms, NR for natural-traffic classification. Capture-the-flag prevention is a distinct endpoint (Supplement S7).
SSH recipient; active probes	LLM Agent Honeypot [P] [54]; possible agent	Prompt compliance and response-time heuristic	Three-month deployment: eight candidate agents among 8,130,731 interactions. No externally verified field labels or detection precision/recall.

missing endpoint: completion checks the requested action, while integrity also checks other orders, shared quotas and the transaction ledger. The writes are deliberately adverse synthetic operations; WRA's research decoys are a separate mechanism.

Defense consequence. An executed-cover design needs a semantic constraint alongside a traffic budget. It can terminate decoys before business execution, or isolate their resources, state and planner feedback. These are testable design choices: measure adapted inference, additional work, original-task completion and collateral state changes on the same protected runs. The published WRA text supports the intervention/utility distinction; the OHTTP refit/cost example depends on one preprint.

5.3. Execution changes delimit the two findings

A private recurring routine can be recognizable while its classifier transfers poorly to a new execution organization. In the communication-graph study, composition holdout keeps the nine workflow labels but changes their capability combinations: accuracy changes from about .67 to .18, with zero recall for seven classes [3, Sec. 9.12]. The cloud benchmark's graph/star change separately reports about .83 to .52 under its corresponding feature protocol [4]. Each axis depends on one preprint.

Implementation and failure handling can also change the observation. Trace reports macro-F1 .981 in its mixed-session model-attribution test and .815 on scaffold hold-out, retaining 1,010 of 1,050 planned clean runs [51]. AgentTrace's active probes reach 66.8% attribution from initial outcomes and 96.0% from ordered repairs at fifty probes [50]. Excluding failed sessions and observing recoverable errors are different operations. These source-specific conditions delimit what an attack or defense says about new workloads; they do not estimate a common filtering effect.

Finally, a traffic target retains its own security meaning. ANT labels benchmark risk, AgentTell tests action choice under a disclosure rule, and CHeaT measures prevention of a defined CTF objective [9], [8], [10]. Authorization additionally requires a principal and policy context. Two refunds that differ only in an unobserved approval can have identical traffic. This conditional argument identifies the extra policy input an authorization monitor needs.

6. Defense Mechanisms and Execution Costs

Defense placement determines both the signal changed and the party that must enforce the change.

TABLE 5. CONDITIONAL RISK MAP. A ROW APPLIES WHEN ITS ACCESS AND EXECUTION CONDITIONS HOLD. THE FINAL COLUMN STATES A PROTECTION PRIORITY OR DESIGN HYPOTHESIS, NOT A DEMONSTRATED MITIGATION. SCORES, TARGETS AND RIGHTS DIFFER ACROSS ROWS, SO THERE IS NO GLOBAL DANGER RANKING.

Interface and signal	Condition and evidenced inference	Protection consequence
Tool egress: selected domains and timing	The observer can associate contacted names with a supplied research trace. WRA recovers query meaning; linked histories use a separate persona endpoint [2].	Prioritize destination association when the private target is research intent. Reduced name visibility needs a residual traffic assessment; executed decoys need task-utility constraints.
Model link: lengths, directions and timing	Model traffic is visible and associated with the evaluated task. ANT and the cloud benchmark measure task information in this view [9], [4]. Initial instructions and later feedback coexist.	Tool-only protection leaves this view to be assessed. Compare model-only and joint adaptive attacks; isolate feedback before choosing a feedback-specific defense.
Coordination: message graph, volume and sequence	Workflow records or coordination traffic expose execution organization. Composition and topology transfer tests change that organization under known task labels [3], [4].	Target observable scheduling and association, not just one payload size. Evaluate changed compositions and topology separately from recurrent-workflow privacy.
Client link: packet counts, directions and bursts	AgentPrint's supplied browser interactions support hosted-agent identity inference [1]. The result presumes those observation units and identities.	Client-facing traffic needs its own protection assessment even if tool links are hidden. Task discovery and new identities require separate tests.
Contacted recipient: headers, commands or arguments	MARK, Trace and PrivacyTrace use recipient-visible representations. In PrivacyTrace, metadata alone is below its prior, while adding arguments improves attribute inference [6], [51], [37].	Minimize sensitive request content and assess implementation signatures. Transport encryption does not conceal the recipient's intended plaintext.
Active probing: challenges, choices and repairs	A party can elicit responses and observe later behavior. AgentTrace attributes models from repairs; AgentTell observes a choice under a disclosure rule [50], [8].	Test error normalization and restrictions on cross-context disclosures while retaining task recovery. These hypotheses need active-access evaluations.

6.1. What a defense changes

Table 6 organizes six intervention choices by enforcement point, trust, cost and residual signal. Association defenses hide a destination or its link to a user. Padding changes sizes, while scheduling changes temporal structure. Transport cover can terminate before business execution; execution cover adds real work. State isolation limits shared-resource signals, and content minimization limits what a recipient learns. These choices can be combined, but their costs and remaining views must be evaluated together.

6.2. Relating communication defenses to Agent execution

STP illustrates transport cover that can terminate at a tunnel endpoint [36]; Walkie-Talkie and FRONT provide shaping mechanisms [17], [18]. For the ticket Agent, this could mean padding the tool response at an egress proxy, pacing model tokens, or sending cover to a trusted terminator. Calling the refund tool as a decoy would instead cross the business-state boundary. The intervention must therefore specify both where cover is removed and what work it can trigger.

The measured design points occupy different parts of this space. WRA changes executed research, live OHTTP changes destination association, and the paired forwarding study changes the context sent to the model. Section 5 gives the source-specific recovery and cost results; Section 7 tests cover execution, forwarding and connection policies. Supplement S7 retains the complete defense map, including identity obfuscation and adversarial-Agent blocking as distinct objectives.

6.3. Choosing a layer in practice

Start from the private object and the observer's rights. Protect destination associations when contacted domains expose the query; protect visible model output sizes and timing when those carry the target; minimize sensitive arguments when the recipient is the observer. Then test what remains on other permitted links. A shared gateway changes which tasks contribute to a connection, so its connection policy belongs in the evaluation. For executed cover, bind privacy, token/tool use, requested-task completion and whole-state correctness to the same protected runs. These choices turn the risk map into a deployment-specific protection plan.

TABLE 6. DEFENSE DESIGN SPACE: ENFORCEMENT AUTHORITY, COSTS TO MEASURE AND REMAINING EXPOSURE. SOURCE-SPECIFIC MEASUREMENTS APPEAR IN THE TEXT AND SUPPLEMENT S7.

Layer / mechanism	What must be trusted or controlled	Costs to measure together	Residual exposure and practical decision
Naming / association: encrypted name resolution, relays, OHTTP	Name resolver or relay placement; the parties able to link the user to the destination	Added hops, bytes, latency and availability	Shape can survive name hiding. Choose this layer when destination association is the private target; then test remaining observers.
Representation: message, TLS-record or packet padding	Component that controls the final visible size; padding survives later encoding	Added bytes, record/packet count, fragmentation	Timing, count and other links remain. State the padding layer and test downstream captures.
Scheduling: batching, pacing, fixed-rate cover	Sender/relay controls output schedule and cover lifetime	First-response delay, completion delay, bandwidth during idle periods	Admission, termination and overflow can reveal activity. Budget these events as part of the visible trace.
Transport cover: dummy exchanges discarded at a tunnel endpoint	Trusted cover terminator; no dispatch into business tools	Cover bandwidth, queueing, resource use	Other endpoints still see their inputs. Prefer this boundary when the aim is traffic shaping without extra business work.
Execution cover: extra research or tool calls	Tool privileges, isolated state and whether decoy feedback reaches the planner	Model tokens, tool quota, task success, state changes and latency	A decoy can affect the real execution. Measure privacy and original-task utility on the same protected runs.
Shared-state isolation / minimization	Tenant separation, cache keys and information passed between services	Lost cache reuse, compute, latency, task accuracy	Isolation removes a sharing channel, not every traffic signal. Minimize tool arguments and feedback according to each recipient's role.

7. Controlled Tests of the Execution Boundaries

Two studies complement the evidence map. The executed-cover study measures the distinction between task completion and business-state integrity. The separate forwarding pilot tests feedback and concurrent observations. Their protocols and uncertainty populations remain distinct.

7.1. Task completion can conceal business-state damage

The study compares six cover policies on matched synthetic support tickets, with real LLM decisions and tool execution. It contains 3,600 episodes: two fixed-weight models, six conditions and 300 fixtures per cell, partitioned into 120 training, 60 validation and 120 test fixtures. Each test class has twenty fixtures. The same fixtures and decoy draws are paired across conditions; request phrasings recur across splits.

Execution and observation. Qwen3:8B and Llama3.1:8B choose checks, business actions, objects, recovery and termination through LangGraph. A controller fetches the ticket and enforces prerequisites without supplying the correct business action. Seven destination-mapped local MCP Streamable HTTP services use actual TLS 1.3/HTTP/2 connections, a common `/mcp` path and request/response padding quanta of 4,096/16,384 bytes. The observer receives destinations and bidirectional ciphertext byte counts within supplied episode and call windows. Model traffic and physical timing are outside this observation.

The six conditions are *no cover*, *proxy discard*, *read cover*, *isolated execution*, *live write*, and *isolated execution with visible responses*. A frozen schedule places one or two random decoys among early tool calls. Proxy discard computes a response on a disposable copy without retaining business effects; isolated execution keeps a separate state clone. Live writes modify the primary synthetic fixture. The visible control forwards clearly marked cover responses from the isolated clone to the planner. Other cover responses are hidden. The decoy writes deliberately exercise collateral effects such as changing another order or consuming a shared quota.

Three endpoints on the same runs. Recognition predicts the requested action even when execution fails. Standardized logistic regression (LR) and a random forest (RF) train within each condition; validation accuracy selects the primary attacker. We also retain both attacks, a frozen baseline-trained attack and destination-position rules. *Goal completion* requires the requested action and object, prerequisites and target postcondition. *State integrity* compares the entire final business state with the single intended-action reference, including other orders, shared resources and ledger entries. A correct final state alone need not satisfy the stricter action/object history required for goal completion.

For each endpoint, five cover-minus-baseline contrasts and one visible-minus-hidden contrast per model form a prespecified twelve-test Holm family. Conditional intervals

resample paired test fixtures by action; they do not include learner-selection or deployment uncertainty. All 3,600 formal episodes have terminal results. Supplement S8 gives the frozen design, all attacks, paired comparisons and state replay.

Completion masks collateral damage. Live-write cover completes 102/120 Qwen tasks and 76/120 Llama tasks, but only 7/120 and 3/120 final states match the intended reference (Table 7). Thus 95 and 73 completed tasks have incorrect global state. Relative to no cover, integrity falls from 116 to 7 and from 69 to 3: paired differences of -90.8 and -55.0 percentage points, with conditional 95% intervals $[-95.0, -85.8]$ and $[-62.5, -47.5]$. Both survive Holm correction. Completion falls by 10.0 and 5.0 points, but only Qwen's contrast is significant (adjusted $p = .022$ versus 1). Llama already has thirteen completed tasks with state errors without cover, so absolute failures cannot all be attributed to decoys.

Privacy and planner feedback remain separate results. All ten primary cover-minus-baseline recognition contrasts survive their Holm correction. Live-write scores, 62/120 and 80/120, sit within the ranges observed for the other cover policies. This descriptive overlap does not establish equivalence. The Qwen last-destination rule reaches 75–77/120 under cover and outperforms the selected learner; substantial residual exposure remains. Cover call bytes cost 1.28–1.35 times the paired baseline.

The hidden/visible isolated arms return identical decoy responses in every test pair. Honest action sequences differ in 16 Qwen pairs and 65 Llama pairs, and visible responses increase mean prompt tokens by 18.1% and 18.9%. Yet all three primary visible-minus-hidden endpoints have adjusted $p = 1$. Hidden arms also exhibit path variation. The control therefore supplies individual execution differences without establishing an average utility or recognition effect.

7.2. Feedback exposure and concurrent execution

The separate forwarding pilot uses identical initial messages, task-bearing tool feedback and randomized session compositions. Six actions require one to three checks, with a controlled subset encountering a recoverable error. *Full* forwards the whole response; *Projected* removes supplementary records while retaining task-required fields. These records deliberately provide a dependency to examine. Projection changes content, encoded length and later execution together.

Two fixed 8B models perform 3,120 local executions over LangGraph, MCP Streamable HTTP and streaming TLS 1.3/HTTP/2. Each model/condition has 156/36/120 train/validation/test tasks in 32/8/24 sessions. Four test sessions are empty. Dedicated and Shared connections permit four workers; Joint adds model-link padding and pacing. Wire checks find overlapping model streams in 107/128 sessions in each shared condition. The local gateway represents a remotely accessed model service on one host. It does not give a network observer access to an in-process model.

First-feedback point differences remain inconclusive. Full first-feedback LR recognition is 43/120 for Qwen and 40/120 for Llama; Projected gives 23/120 and 22/120. Prefix scores are 25/120 and 19/120. All eight local primary LR/k-FP contrasts exceed .05 after exact Holm correction, including every post-feedback–prefix comparison; the minimum is .0508. Exploratory RF gives 50 versus 30 and 59 versus 19. Complete model-view LR instead changes from 46 to 44 and from 62 to 33; RF changes from 61 to 31 and from 73 to 34. The complete execution retains task fields and model responses, so its exposure is distinct from the first-feedback contrast.

Each local Full/Projected test condition has 518 tool calls, with equal counts in every pair but different call order or actions in 35 Qwen and 26 Llama pairs. The same-count observation excludes fewer calls as the explanation of Llama's decline; it does not isolate feedback content from length and order. The separate hosted-alias retest has 192 attempts and no adjusted primary rejection. It changes network path and backend together. Supplement S1 reports every LR, k-FP, RF and CNN result, all phases, the hosted outcomes and conditional refit intervals.

Connection sharing needs event-level evaluation. Concurrent supplied-window features pool visible connections during overlapping tasks. Serial-to-concurrent recognition changes therefore combine connection policy and window mixing. Continuous LR/RF detectors instead receive session boundaries and connection identities, without task windows or private stream assignments. Under Shared at IoU .5, instance recall is 35.0–43.3%, precision 4.3–18.6%, and false events 8.21–44.79 per test session. At IoU .3, all-active exceeds both learned Shared detectors in F1 for both models; at .5 the ordering reverses. The supplement reports .3/.5/.7 and empty-session controls. These results diagnose the detectors and overlap rule, not general deployment difficulty.

Held-out goal completion is 119/120 for Qwen in all conditions and 117/120 for Llama except Projected at 118/120. Projection costs about .78 of Full's model bytes. Joint shaping costs 19.5 and 22.5 times Shared's model bytes, while model-only LR changes from 25 to 18 and from 19 to 25. These mixed results supply measured design points, with no established end-to-end protection advantage.

8. Evidence Availability and Reporting

Attack adaptation and original-task utility occur in different mapped protection settings. Table 8 identifies four eligible query/workflow privacy comparisons: two document adaptation and one documents matched original-task utility; none locates both. The adapted settings share one preprint, while the utility observation is WRA's published visibility comparison.

Different objectives need different denominators. Three identity-obfuscation settings, two adversarial-Agent blocking settings and thirty-five without a protection comparison remain outside this denominator. The access map

TABLE 7. EXECUTED COVER: ALL HELD-OUT CONDITIONS, 120 PAIRED FIXTURES PER MODEL AND CONDITION. RECOGNITION COLUMNS COUNT CORRECT REQUESTED-ACTION PREDICTIONS; CHANCE IS 20/120. THE LEARNER IS SELECTED USING VALIDATION LR/RF ACCURACY. THE POSITION RULE REMAINS VISIBLE WHEN IT IS STRONGER. GOAL, WHOLE-STATE INTEGRITY AND COMPLETED TASKS WITH STATE ERRORS ARE SEPARATE ENDPOINTS.

Model	Cover policy	Selected	Last dest.	Goal	State	Goal + error	Call bytes ×
Qwen3:8B	No cover	118	116	114	116	0	1.00
Qwen3:8B	Proxy discard	62	77	114	116	0	1.34
Qwen3:8B	Read cover	62	77	114	115	0	1.34
Qwen3:8B	Isolated, hidden	61	77	114	115	0	1.34
Qwen3:8B	Live write	62	75	102	7	95	1.34
Qwen3:8B	Isolated, visible	63	77	113	114	1	1.35
Llama3.1:8B	No cover	113	79	82	69	13	1.00
Llama3.1:8B	Proxy discard	79	67	80	68	13	1.29
Llama3.1:8B	Read cover	72	67	84	71	14	1.28
Llama3.1:8B	Isolated, hidden	82	67	83	70	13	1.28
Llama3.1:8B	Live write	80	65	76	3	73	1.29
Llama3.1:8B	Isolated, visible	81	68	83	73	11	1.29

TABLE 8. EVIDENCE LOCATED IN THE RETAINED RECORDS. COUNTS USE ELIGIBLE SETTINGS; THE SECOND COLUMN ALSO GIVES DISTINCT SOURCES. "NOT LOCATED" LEAVES EVIDENCE UNRESOLVED WITHIN THE INSPECTED SCOPE. THESE COUNTS DESCRIBE AVAILABILITY, NOT STUDY QUALITY OR CHECKLIST VALIDITY.

Question	Eligible settings / sources	Located	Supporting settings and consequence
Protected-condition attack adaptation	4 / 2	2/4	Communication graphs: transformed captures and live OHTTP. Both come from one preprint.
Original-task utility under the same intervention	4 / 2	1/4	WRA visibility comparison. It does not price WRA's executed-decoy condition.
Adaptation and original-task utility in one setting	4 / 2	0/4	Adaptation and utility are located in different records; transport cost alone does not fill the task-outcome field.
Original-task utility for executed cover	1 / 1	0/1	WRA's five research decoys report recovery and extra research; matched original-task utility is not located.

separately counts twenty-two transit settings, ten recipient settings, eleven active settings and one downstream preference analysis. These author-assigned classes partition settings; source groups overlap. The counts describe the retained corpus. Agreement in the AI manual pilot does not validate these assignments independently. Failure/outcome availability is shown as encoding sensitivity in Supplement S4.1, with each encoding's own applicability denominator, rather than a single settled coverage rate.

A compact reporting contract. Arp et al. and WF guidelines establish sampling, labels, observation, splits and adaptive evaluation [55], [15], [16]. Our eight-field checklist adds service/access mapping, label stage and original-task costs. The supplementary reporting section provides the checkable form, guideline correspondence, eligibility ledger and completed WRA, OHTTP and ANT examples. Each entry records evidence, a missing item or inapplicability.

The reporting unit is the intervention: which observer remains, how the attack adapts, and what happens to the original task and surrounding state. Recording these together would close the mapped privacy–utility gap.

9. Synthesis and Research Agenda

9.1. Source dependencies of the two findings

The transfer matrix separates established conditions from evidence for their Agent application. Model- and tool-side task observations draw on several studies; the decoy/utility mismatch comes from WRA's published text. Live OHTTP refitting and each composition, topology or scaffold transfer axis depend on individual preprints. The two located adapted privacy settings share one source. These dependencies matter more than treating every setting as an independent confirmation.

Source withdrawal tracks thirty-three statements across seven comparisons. The source appendix gives the complete summary and twenty removal scenarios, including individual sources, shared-author groups and publication filtering.

The strict published-text subset retains WRA and CHeaT: query recovery, the decoy/visibility distinction and a defined prevention outcome survive. Other label, population and policy contrasts depend on the wider corpus. Removing one transfer source removes evidence for that axis, rather than changing the other axes. The two local studies add separate controlled evidence. The feedback pilot

remains inconclusive after correction. The executed-cover study supports a narrower positive finding: under its adverse-write policy, successful tasks can leave damaged business state. Neither changes the number of literature sources or replicates WRA's decoy mechanism.

9.2. Tests that would change the protection decision

The next studies should measure the information and execution that cross a defense boundary. Table 9 gives four concrete comparisons; the supplementary agenda supplies their source basis. The feedback intervention tests one controlled path. The executed-cover study tests one semantic boundary; arbitrary quotas, legitimate concurrent updates and production tools require application-specific state tolerances. Longer histories and different gateways can also change the observation.

For practitioners, three decisions follow. First, map a private target to the model, tool and recipient observations actually available to the adversary. Second, check adapted joint and permitted single-link attacks after shaping or forwarding changes. Third, if cover executes, specify which task resources, state and feedback it may affect, then measure those quantities on the protected runs. A design that lowers inference while meeting these application constraints would add a useful point to the sparse defense map.

10. Validity, Ethics and Availability

Evidence scope. Source locators and withdrawal analysis expose dependencies; index failures and query misses limit coverage. Extraction is author-performed with AI assistance. The two AI recodings are non-blind and dependent; their agreement is a coding-manual-pilot diagnostic, not independent human reliability. The source-specific measurements and our controlled tasks represent different populations.

The forwarding pilot covers six synthetic actions with deliberately category-bearing supplementary records on one host. All its adjusted primary tests exceed .05. The forwarding intervention changes content, encoded length and execution order together; equal tool-call counts do not isolate the first two. Complete-execution and overlap-threshold contrasts are reported alongside the focal phase results. Session resampling, including LR refitting, measures variation within this workload and selected learner family. Recognition receives phase or task windows; localization receives session boundaries and synthetic live background traffic. These conditions leave generalization to other workloads, enterprise background and gateways open. The remote retest changes the backend and network together and uses an alias without pinned weights.

The executed-cover study separately uses 120 unique paired test fixtures, reused phrasings, local destinations and supplied call windows. Its intervals condition on fitted attacks and this constructed workload. Live writes deliberately modify business state; their measured frequency and severity do not estimate production cover harms. The isolated visible-response endpoint differences remain nonsignificant. The weaker Llama baseline and all terminal failures remain in the denominators.

Earlier serial, common-prefix and fixed-composition studies are archived as separate populations in the review artifact, with their failures, inconclusive results and unknown responses. Their outcomes are not pooled with the present experiment.

Ethics and assistance. Inputs are synthetic, business actions update isolated fixtures, and capture filters restrict collection to the experiment. Remote model use has the account holder's authorization; provider-specific research permission and applicable terms remain unverified. Known failures and unknown responses are retained separately. No human-participant experiment, customer operation or hosted-provider vulnerability finding enters the study. AI tools assisted source extraction, implementation and writing; saved observations and executable analyses support the reported measurements.

Traffic inference can enable unwanted profiling as well as monitoring. The review package contains source-extraction records, synthetic metadata, features, predictions and offline analyses. The executed-cover artifact additionally includes synthetic fixture and planner records sufficient to replay state outcomes; these contain no customer data. Credentials, private user transcripts, raw packet captures and third-party full texts are excluded. Reusable analysis methods still carry misuse potential. A protection summary should report adapted attacks, remaining views and original-task outcomes together; a frozen attack's decline alone can overstate the measured protection.

Availability. The accompanying review package connects the forty-four-setting evidence matrix, transfer relations, eligibility ledger, source locators and dependency calculations to the claims. Synthetic metadata, predictions, outcome markers, uncertainty scripts and manifests provide an offline verification route. Raw-to-metadata reconstruction needs the excluded captures; the mutable remote alias cannot reproduce fixed weights. The supplement presents the current study and source basis; the artifact also preserves the earlier experimental reports and all saved measurements. Compilable LaTeX and both PDFs accompany the package.

11. Conclusion

Classical traffic-analysis knowledge carries into Agent security through explicit access, observation and intervention conditions. The transfer matrix identifies two execution boundaries requiring additional checks: feedback crosses separately protected services, and executed cover changes application work. The fifteen-source evidence map ties these boundaries to actual observation rights and measurements. Among four query/workflow privacy settings, adaptation and original-task utility are located in different records; the published executed-decoy result leaves matched task utility open. The LLM-planned cover study shows the consequence of an incomplete utility endpoint: 95 and 73 completed tasks still have incorrect state under live writes. Both integrity

TABLE 9. TESTABLE OPEN QUESTIONS DERIVED FROM THE SYNTHESIS. Q1 SPECIFIES TRANSFER TESTS. Q2 SEPARATES THE FORWARDING MEDIATORS; Q3 EXTENDS THE CONTROLLED COVER STUDY TO APPLICATION CONSTRAINTS; Q4 EXTENDS LIVE DETECTION TO NEW WORKLOADS AND DEPLOYMENTS. EACH GIVES A RESULT THAT COULD SUPPORT OR REVISE THE PROPOSED EXPLANATION.

Question and unresolved mechanism	Controlled comparison	Discriminating result
Q1. Which signals survive execution changes? Do attacks identify recurring task structure or features tied to its implementation?	Keep task labels and observation rights fixed. Hold out compositions, coordination and client/scaffold versions separately, then jointly. Compare destination, volume/timing, sequence and recipient-visible feature families under declared frozen and adapted training.	Identify a feature family whose inference survives specified changes, or a failure tied to one changed condition. Report per-axis and joint transfer with attempted-run coverage; a pooled average cannot establish invariance.
Q2. How does feedback reach model traffic? Which part is content, length or later action structure?	Fix initial instructions and task-required fields. Pair length-matched neutral padding with validity-preserving replacement of supplementary content; hold encoding and follow-up policy fixed where possible. Vary record count separately. Compare prefix, first-feedback and complete model views under tool-only and joint protection.	Additional held-out inference after feedback, with controls that exclude the competing path, supports that attribution. Report an effect interval against a prespecified meaningful-effect threshold. A nonsignificant difference alone does not rule out leakage.
Q3. Can cover preserve task semantics? When should it stop at transport or execute in isolation?	Extend the six-condition LLM case to production tool contracts and legitimate concurrent updates. Compare no cover, transport termination and isolated execution under declared resource budgets. Refit permitted attackers; measure bytes, delay, original-task completion, quota use, state changes and feedback entering the planner.	A privacy–utility operating point satisfying prespecified state and task tolerances supports the design. Resource exhaustion, changed business state or impaired completion identifies the constraint it violates.
Q4. Which inference survives task discovery? Does cross-link association recover information hidden by waits and multiplexing?	Train on one concurrency/load and test new compositions, loads and gateways. Compare tool-only and associated tool/model views with observable flow identifiers; keep ground-truth task and stream mappings outside attacker inputs. Include real unrelated traffic and constant-output detector references.	Measure interval recall, false alarms per unit time and correctly localized inference together. Gains must survive changed workloads and loads at a stated false-alarm budget.

losses survive correction, while visible-decoy effects and the separate feedback pilot remain inconclusive. The practical priorities are to evaluate every permitted residual view and to measure privacy, resource use, task completion and collateral state changes on the same protected executions.

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